Introduction to Machine Learning

Session 1 — Lecture notes

Machine Learning → Fundamental for AI

Machine Learning is the subset of artificial intelligence focused on algorithms that can learn the patterns of training data and, subsequently, make accurate inferences about new data.

It is the backbone of most modern AI systems (forecasting models, autonomous vehicles, LLMs, and generative AI).

Classical (symbolic) AI encodes human knowledge and logic explicitly.

ML (in its infancy) dealt with two problems:

- Classification
- Regression

What we call a model is its architecture and parameters.

Perceptron

1957, Cornell University (Ithaca, NY) — Frank Rosenblatt

Two classes that are linearly separable.

Process for 2D dataset:

Step 1

Parameters: $w = [w_1, w_2], b$

Data: $x_i = [x_{i1}, x_{i2}]$

Estimated class: $\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + b$

$w_1x_1 + w_2x_2 + b = 0$ is the decision boundary. Choosing the weights is choosing the boundary!

Step 2

if $\hat{y} \geq 0 \rightarrow class = 1$

else $\rightarrow class = 0$

Training

Iterative process, in general:

$w += w + \eta(y - \hat{y})x$

η = learning rate,

y = true class,

ŷ = prediction

If the prediction is correct, there is no 'nudge' — as this will be 0. The 'nudge' impacts both translation & rotation. If the dataset is linearly separable, the perceptron training guarantees success.

Now, the detailed update process:

$$w_{1\,new} = w_{1\,old} + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i) x_{1i}$$

$$w_{2\,new} = w_{2\,old} + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i) x_{2i}$$

$$b_{new} = b_{old} + \eta \sum_{i=1}^{n} (y_i - \hat{y}_i)$$

Evaluation

Dataset Division

A model that memorises its training data can score perfectly on it while being useless on anything new.

Three data types:

Training	Validation	Test
0.6	0.2	0.2
Model selection	Tuning / threshold choice	Run exactly once! Results / metrics

Confusion Matrix

	Predicted: Positives	Predicted: Negatives
Actual: Positives	TP	FN
Actual: Negatives	FP	TN

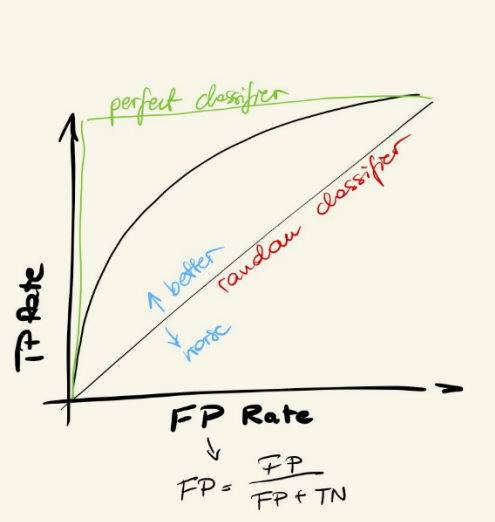
- True Positive
- True Negative
- False Positive — predicted True, was Negative
- False Negative — prediction Negative, was True

Precision & Recall (Metrics)

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

F1 Score — Harmonic Mean of Precision & Recall



$$F1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

ROC Curve

Receiver Operating Characteristic — plotted for different thresholds.

Area under the curve:

- 1 → perfect separation
- 0.5 → random (uninformative)